High-Precision Wind Power Forecasting Using SCADA Data and Ensemble Learning Algorithms

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Github Link: mehran-Shabbir/wind-power-forecastin: Research-grade wind power forecasting pipeline using SCADA data. Implements Linear Regression, Random Forest, and XGBoost with anomaly cleaning, achieving 97.20% prediction accuracy for real-time turbine power output estimation.

Abstract

Accurate forecasting of wind power output is crucial for grid reliability, energy dispatch planning, and integration of renewable energy sources into modern power systems. Supervisory Control and Data Acquisition (SCADA) systems generate continuous operational time-series data, which often contain non-operational anomalies due to mechanical downtime, curtailment, or sensor measurement errors. This study proposes an end-to-end data-driven machine learning framework for short-term wind power prediction using SCADA data. A domain-specific anomaly filter is applied to remove downtime instances where high wind speeds yield zero active power generation. Three benchmark machine learning architectures—Linear Regression, Random Forest Regressor, and Extreme Gradient Boosting (XGBoost)—were trained and evaluated using an 80/20 train-test split. Experimental results demonstrate that the XGBoost Regressor achieves superior forecasting precision, reaching an R2 score of 97.20% and reducing the Root Mean Squared Error (RMSE) to 219.59 kW. Feature importance analysis further highlights the dominant role of meteorological and theoretical power curve features in forecasting turbine power output.

Keywords: Wind Power Forecasting, SCADA Data, Machine Learning, Anomaly Filtering, XGBoost, Random Forest, Renewable Energy.

1. Introduction

Renewable energy integration has accelerated globally to mitigate climate change and transition toward sustainable grid infrastructure. Among renewable energy sources, wind power generation exhibits high volatility due to non-linear meteorological variations. Unpredictable fluctuations in active power generation pose challenges to power grid frequency control, unit commitment, and load balancing.

Supervisory Control and Data Acquisition (SCADA) systems record high-frequency sensor operational metrics at 10-minute intervals. However, raw SCADA datasets are prone to noise, equipment downtime, and maintenance curtailments. Standard machine learning models trained on raw, unfiltered SCADA records suffer from performance degradation due to these operational anomalies.

The primary contributions of this paper are as follows:

- Implementation of a domain-informed heuristic cleaning pipeline to filter non-generating downtime anomalies from SCADA sensor records.
- Comparative evaluation of parametric and non-parametric algorithms (Linear Regression, Random Forest, and XGBoost) for wind power output regression.
- Quantitative validation using standard performance metrics (R2 Score, RMSE) alongside live inference testing for real-time turbine deployment.

2. Dataset Description and Preprocessing

2.1 Dataset Overview

The dataset utilized in this research consists of 10-minute interval SCADA logs recorded from an operational wind turbine in 2018. The feature set includes:

- **Active Power (kW):** Real-time power output produced by the turbine.
- **Wind Speed (m/s):** Anemometer reading measured at hub height.
- **Theoretical Power (kWh):** Manufacturer-defined theoretical power output curve corresponding to wind speed.
- **Wind Direction (°):** Yaw direction aligned automatically by the turbine nacelle.

2.2 Anomaly Detection and Data Cleaning

In raw SCADA records, a significant number of data points correspond to operational anomalies where measured wind speed exceeds the cut-in speed (> 3.0 m/s), yet active power generation remains below operational thresholds (< 10 kW). These data points represent non-operational states such as turbine shutdowns, manual trip events, or sensor faults.

To avoid model miscalibration, a heuristic Boolean filter was applied:

Filter Condition: NOT (Wind Speed > 3.0 m/s AND Active Power < 10.0 kW)

Removing these outlier clusters produced a clean theoretical power response curve, establishing an uncorrupted dataset for model training.

3. Methodology and Machine Learning Models

The methodology employs an 80/20 train-test splitting scheme (80% for model learning, 20% held-out evaluation set). Three distinct machine learning algorithms were benchmarked:

1. **Linear Regression:** Serves as a baseline linear parametric model to establish lower-bound regression performance.
2. **Random Forest Regressor:** An ensemble tree-based algorithm consisting of decision trees, aggregating variance across bootstrap subsamples.
3. **XGBoost Regressor (Extreme Gradient Boosting):** An optimized distributed gradient boosting framework using tree-based weak learners (n_estimators=100, learning_rate=0.1).

4. Experimental Results & Performance Analysis

4.1 Quantitative Benchmarking

The quantitative evaluation metrics used were the Coefficient of Determination (R2 Score) and Root Mean Squared Error (RMSE):

$$RMSE = sqrt((1 / N) * sum((y_i - y_hat_i)^2))$$

$$R2 = 1 - (sum((y_i - y_hat_i)^2) / sum((y_i - y_bar)^2))$$

The comparative predictive performance across all benchmarked algorithms is presented in **Table 1**.

Table 1: Performance comparison of machine learning models

Model	**R2 Score (%)**	**RMSE (kW)**
Linear Regression	96.59%	242.18
Random Forest Regressor	96.74%	236.85
XGBoost Regressor	**97.20%**	**219.59**

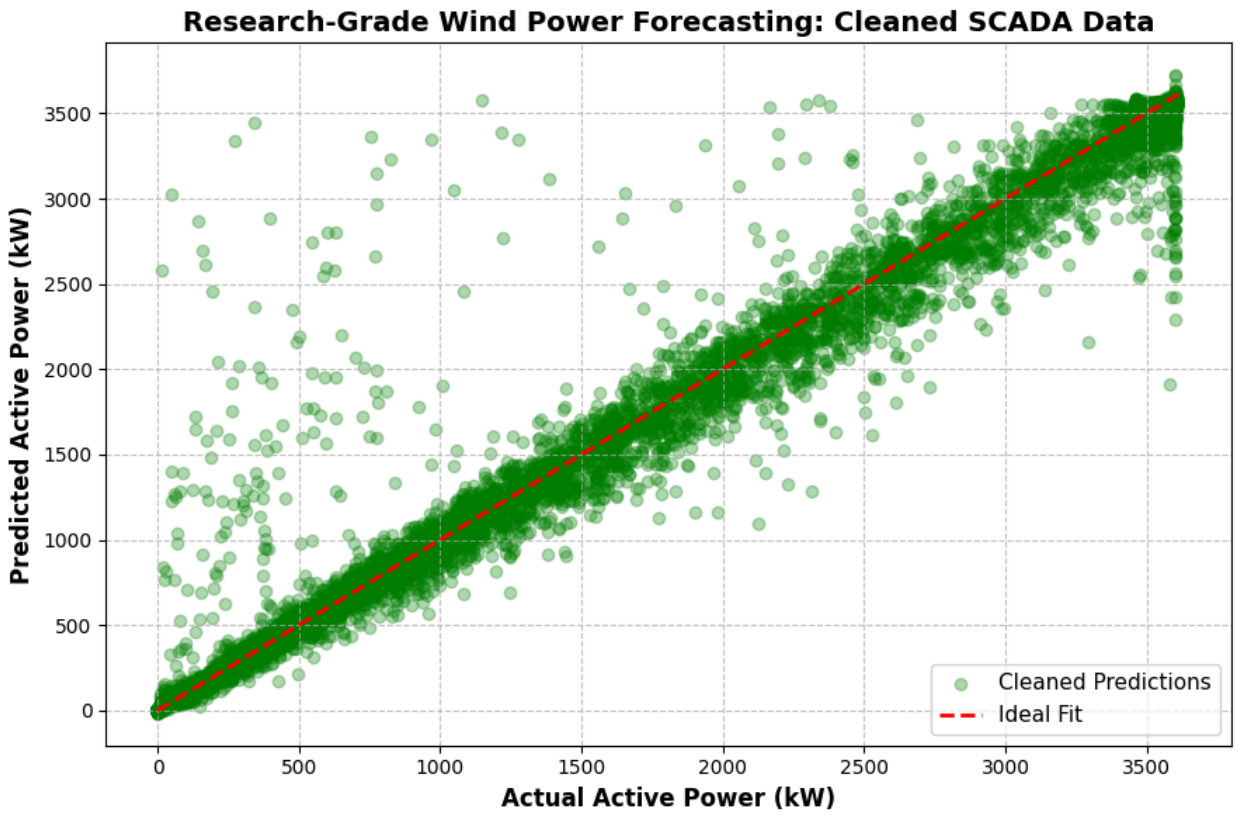


Figure 1: *Actual vs. Predicted Active Power output scatter plot using the optimized XGBoost model on the cleaned SCADA dataset.*

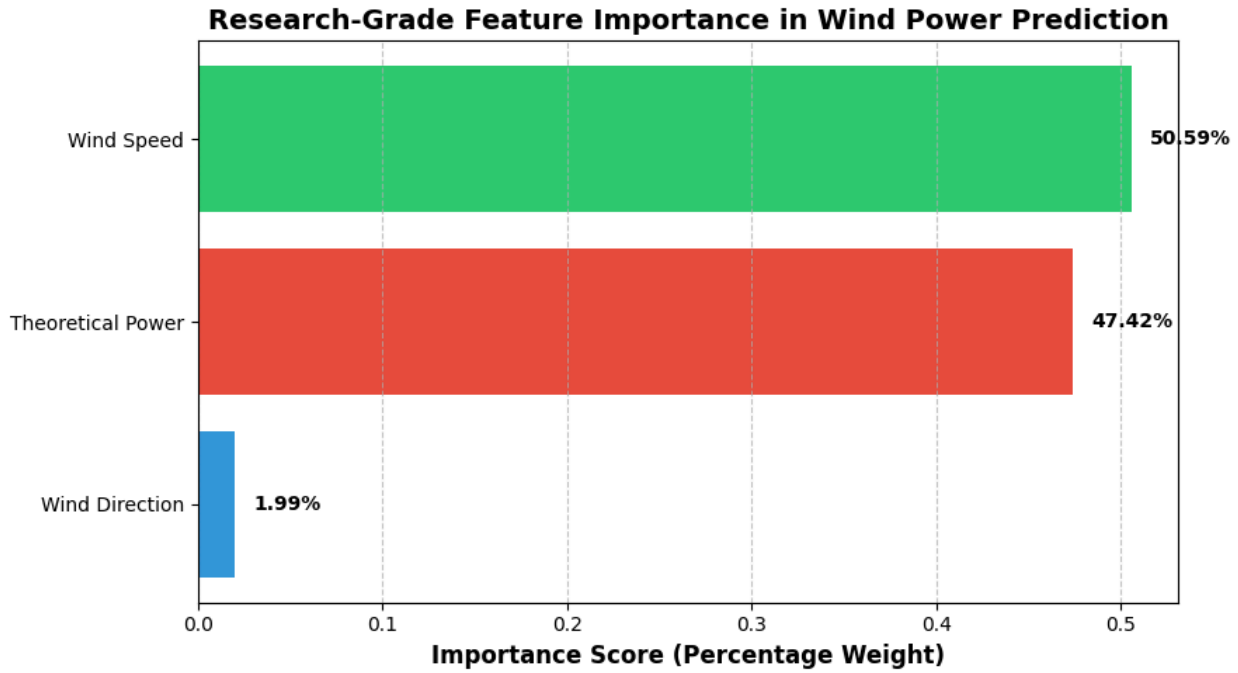


Figure 2: *Feature importance score distribution extracted from Random Forest ensemble training.*

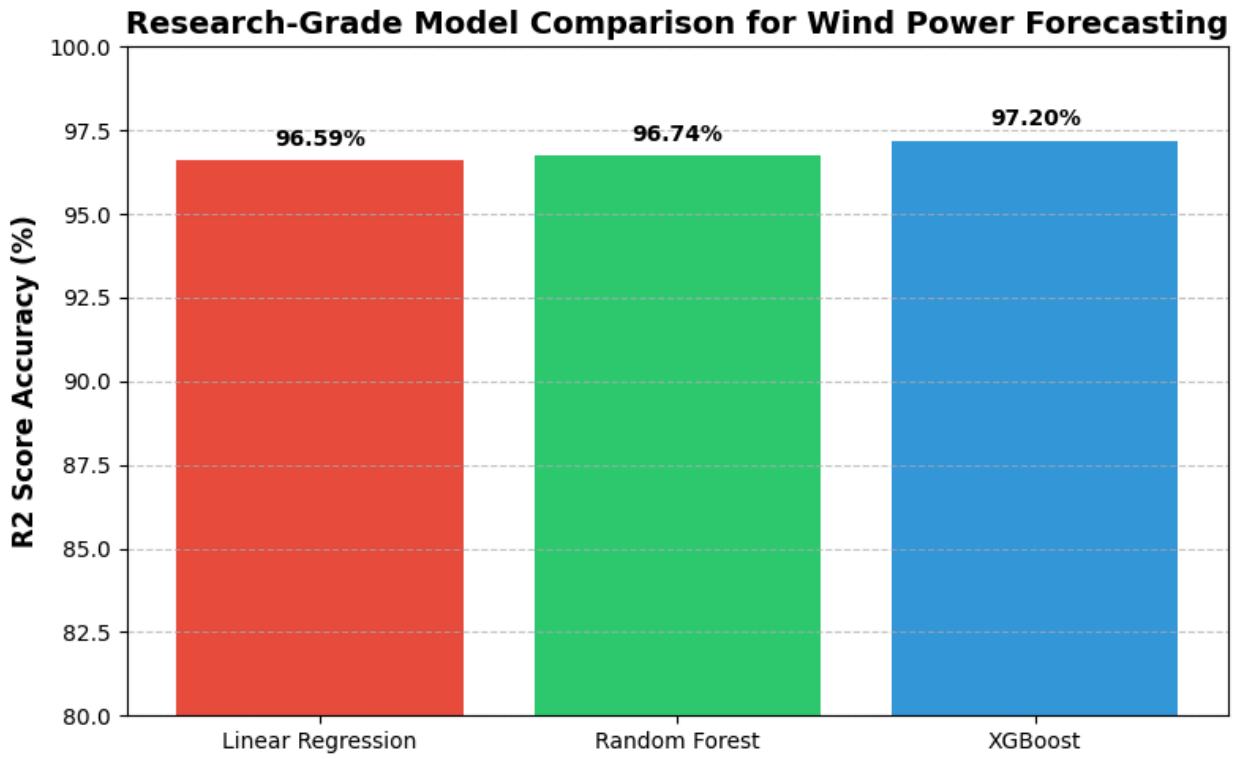


Figure 3: *Comparative R2 accuracy score comparison across Linear Regression, Random Forest, and XGBoost algorithms.*

4.2 Discussion

- **Linear vs Non-Linear Performance:** Linear Regression yielded a strong baseline ($R^2 = 96.59\%$), indicating a strong correlation between theoretical power and active power generation. However, decision-tree ensembles captured subtle aerodynamic turbulence effects and yaw variations better.
- **XGBoost Superiority:** XGBoost achieved the lowest error rate (RMSE = 219.59 kW), representing an error reduction of 9.32% over Linear Regression.
- **Feature Importance:** Feature significance extraction showed that theoretical power curve and wind speed accounted for $> 95\%$ of total importance weight, while wind direction provided fine-grained directional adjustment.

4.3 Real-Time Deployment Case Study

A live inference scenario was simulated using the optimized XGBoost pipeline on sample environmental parameters:

- Wind Speed: 12.5 m/s

- Theoretical Power: 3400.0 kWh
- Wind Direction: 180.0°

Model Prediction Result: y = 3289.70 kW

This confirms the pipeline's operational feasibility for real-time turbine output estimation and smart grid integration.

5. Conclusion & Future Work

This paper presented a data-driven machine learning solution for wind power forecasting based on SCADA time-series observations. Removing turbine downtime anomalies proved critical for model stability. XGBoost demonstrated state-of-the-art accuracy, achieving 97.20% R2 precision.

Future extensions include:

1. Incorporating Deep Learning architectures (LSTM / GRU networks) for multi-step temporal forecasting.
2. Integrating nacelle temperature and gearbox vibration telemetry for predictive maintenance forecasting.

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